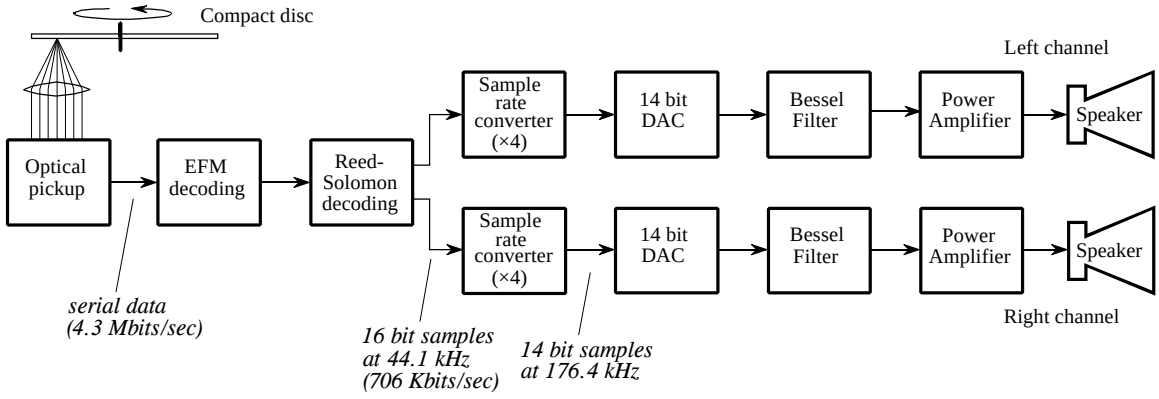

FIGURE 22-6
Compact disc playback block diagram. The digital information is retrieved from the disc with an optical sensor, corrected for EFM and Reed-Solomon encoding, and converted to stereo analog signals.

In addition to EFM, the data are encoded in a format called **two-level Reed-Solomon coding**. This involves combining the left and right stereo channels along with data for error detection and correction. Digital errors detected during playback are either: *corrected* by using the redundant data in the encoding scheme, *concealed* by interpolating between adjacent samples, or *muted* by setting the sample value to zero. These encoding schemes result in the data rate being *tripled*, i.e., 1.4 Mbits/sec for the stereo audio signals versus 4.3 Mbits/sec stored on the disc.

After decoding and error correction, the audio signals are represented as 16 bit samples at a 44.1 kHz sampling rate. In the simplest system, these signals could be run through a 16 bit DAC, followed by a low-pass analog filter. However, this would require high performance analog electronics to pass frequencies below 20 kHz, while rejecting all frequencies above 22.05 kHz, ½ of the sampling rate. A more common method is to use a **multirate** technique, that is, convert the digital data to a higher sampling rate before the DAC. A factor of four is commonly used, converting from 44.1 kHz to 176.4 kHz. This is called **interpolation**, and can be explained as a two step process (although it may not actually be carried out this way). First, three samples with a value of zero are placed between the original samples, producing the higher sampling rate. In the frequency domain, this has the effect of duplicating the 0 to 22.05 kHz spectrum three times, at 22.05 to 44.1 kHz, 41 to 66.15 kHz, and 66.15 to 88.2 kHz. In the second step, an efficient *digital* filter is used to remove the newly added frequencies.

The sample rate increase makes the sampling interval smaller, resulting in a smoother signal being generated by the DAC. The signal still contains frequencies between 20 Hz and 20 kHz; however, the Nyquist frequency has been increased by a factor of four. This means that the analog filter only needs to pass frequencies below 20 kHz, while blocking frequencies above 88.2 kHz. This is usually done with a three pole Bessel filter. Why use a *Bessel* filter if the ear is insensitive to phase? Overkill, remember?